

Justification Doc

Architecture chosen:

The system is split into multiple smaller services that can be deployed independent of each other. These services are

- 1) Claim Management
- 2) Baggage Scan Matching
- 3) Compensation
- 4) Identity Verification
- 5) Notification

Each of these services is responsible for a different part of the domain (like claim registration, cross-terminal scan reconciliation, SLA-driven payouts, passenger identity checks, and status alerts) and communicating over well-defined REST/event interfaces rather than sharing a single codebase or database.

Justification

1. To allow scaling of each service independently even the load on the services are unbalanced or uneven:

FR-001 and NFR-002 address the disruptions during peak travel times. This means that when the Baggage Scan Matching Service faces a spike in cross-terminal lookups, the Claim Management and Notification see relatively steady traffic. Splitting the microservices lets the Scan Matching Service be scaled out on its own during a disruption without over-provisioning the rest of the system.

2. To localize/ isolate any faults from affecting critical workflows:

FR-004 needs an automated trigger tied to a regulatory SLA. Running Compensation as a separate service means a failure or slowdown there cannot block passengers from registering claims (FR-002) or checking live status (FR-003), which is essential to meeting the 99.9% uptime target in NFR-002.

Security aspect

Each service is protected by its own authenticated API and holds only the information it needs. This means that the Notification Service doesn't have access to the tag and flight detail records that belong to the Claim Management. This limits the extent of the damage of a failed service and also limits the exposure of passenger PII referenced in NFR-001, since any single failing service doesn't harm the entire dataset.

Performance

The Baggage Scan Matching Service is deployed and optimized independently, because of this, it can have dedicated resources and its own caching layer calibrated specifically for scan history lookups, which aids NFR-001's under 200ms latency requirement without interference from other independent workflows.